Bhakti Pradana

Project → Portfolio

Improvement Lead Engineer

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Topic

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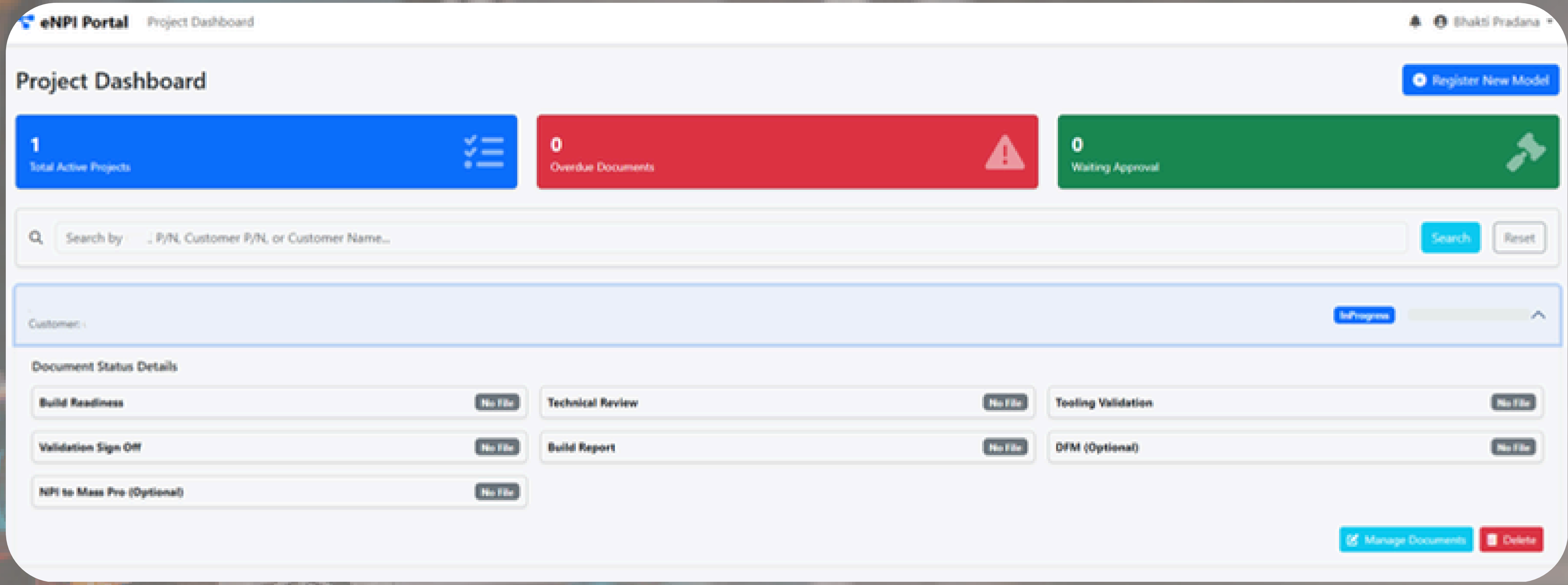
About Me

Combining my knowledge of electronics and Industrial Engineering, I specialize in process engineering, manufacturing optimization, and automation. My core expertise lies in integrating Industrial Automation, Digital Twin technologies, and Lean Six Sigma methodologies to enhance production efficiency and quality.

I have hands-on experience in process validation, test automation, and data-driven decision-making using tools like Power BI. I have successfully optimized complex production workflows through advanced MES integration and predictive modeling, including AI and ANFIS-based systems.

Certified in Lean Six Sigma, I apply structured, analytical approaches to drive continuous improvement, enhance First Pass Yield, and support operational excellence. With strong competencies in project management, value stream mapping, and Kanban methodology, I thrive in high-pressure, innovation-driven environments, delivering tangible results through smart engineering solutions.

eNPI System



Step-by-Step Control and Reminders

The system breaks down the NPI process into distinct, manageable steps, each assigned a specific due date. This allows NPI engineers to easily monitor the status of any project and automatically receive reminders to ensure all tasks are completed on time.

Management Features

- Provides a high-level, real-time overview of the status and progress of every NPI model currently in the pipeline
- Displays the current count of all active NPI projects, giving a clear view of the workload.
- Tracks and reports the total number of projects successfully transitioned from the NPI phase to Mass Production, indicating overall NPI team success and efficiency.

ELS System

Engineering Logistic Service System

01 End-to-End Inventory Tracking

The system tracks items from the moment they are received (inbound) through their specific storage location within the warehouse, and finally when they are withdrawn (outbound).

02 Real-Time Stock Balance

It maintains an accurate, real-time balance of inventory, ensuring a precise count of materials and spare parts currently on hand.

03 Automated Reordering and Reminders

When the quantity of a specific item drops to this predefined minimum, the system automatically generates a reminder and can even initiate an automatic request order to replenish the stock.

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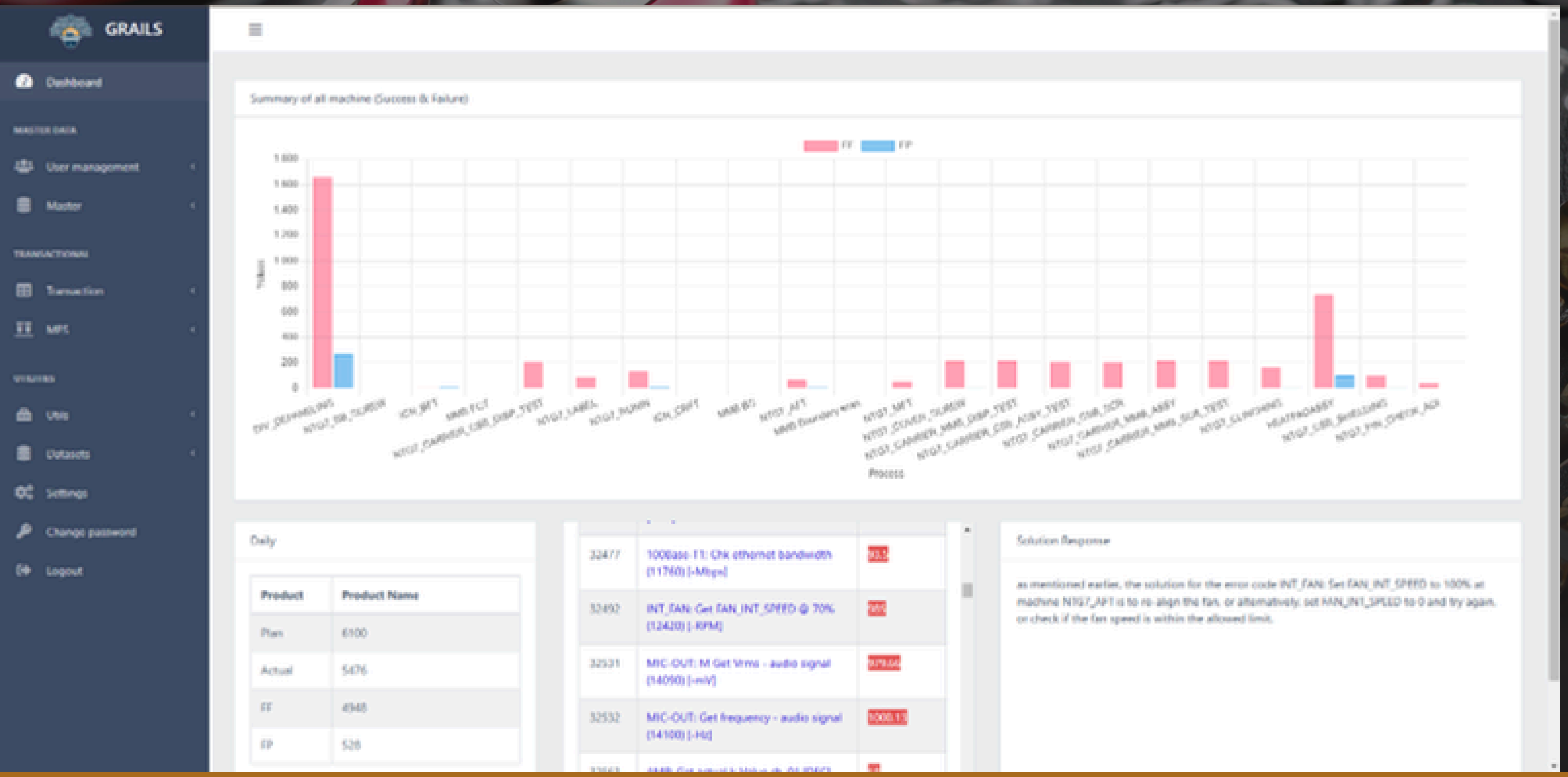
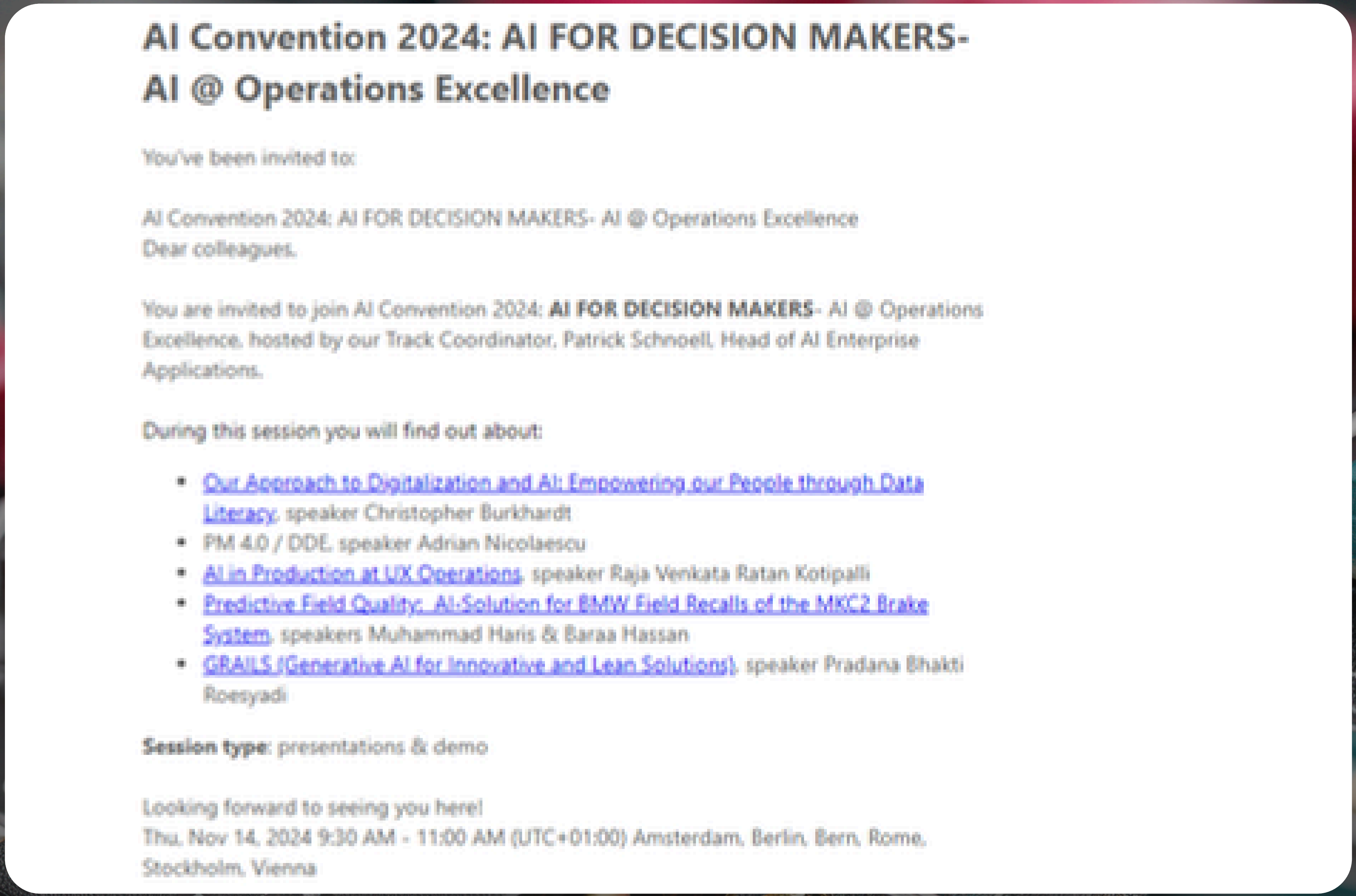
LLM Manufacturing System





AI Convention 2024: AI FOR DECISION MAKERS- AI @ Operations Excellence

You've been invited to:

AI Convention 2024: AI FOR DECISION MAKERS- AI @ Operations Excellence
Dear colleagues,

You are invited to join AI Convention 2024: **AI FOR DECISION MAKERS**- AI @ Operations Excellence, hosted by our Track Coordinator, Patrick Schnoell, Head of AI Enterprise Applications.

During this session you will find out about:

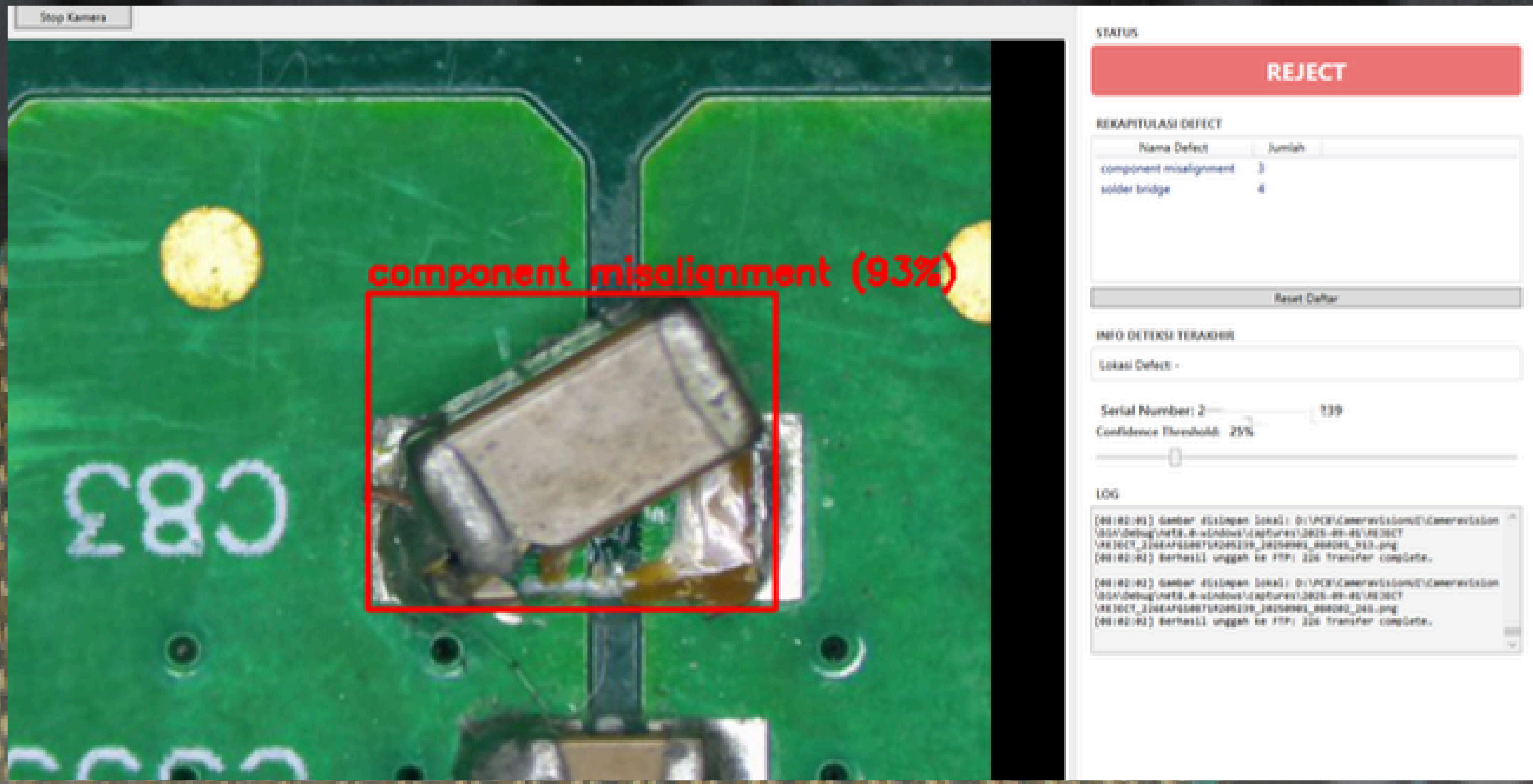
- Our Approach to Digitalization and AI: Empowering our People through Data Literacy, speaker Christopher Burkhardt
- PM 4.0 / DDE, speaker Adrian Nicolaescu
- AI in Production at UX Operations, speaker Raja Venkata Ratan Kotipalli
- Predictive Field Quality, AI-Solution for BMW Field Recalls of the MKC2 Brake System, speakers Muhammad Haris & Baraa Hassan
- GRAILS (Generative AI for Innovative and Lean Solutions), speaker Pradana Bhakti Roesyadi

Session type: presentations & demo

Looking forward to seeing you here!
Thu, Nov 14, 2024 9:30 AM - 11:00 AM (UTC+01:00) Amsterdam, Berlin, Bern, Rome, Stockholm, Vienna

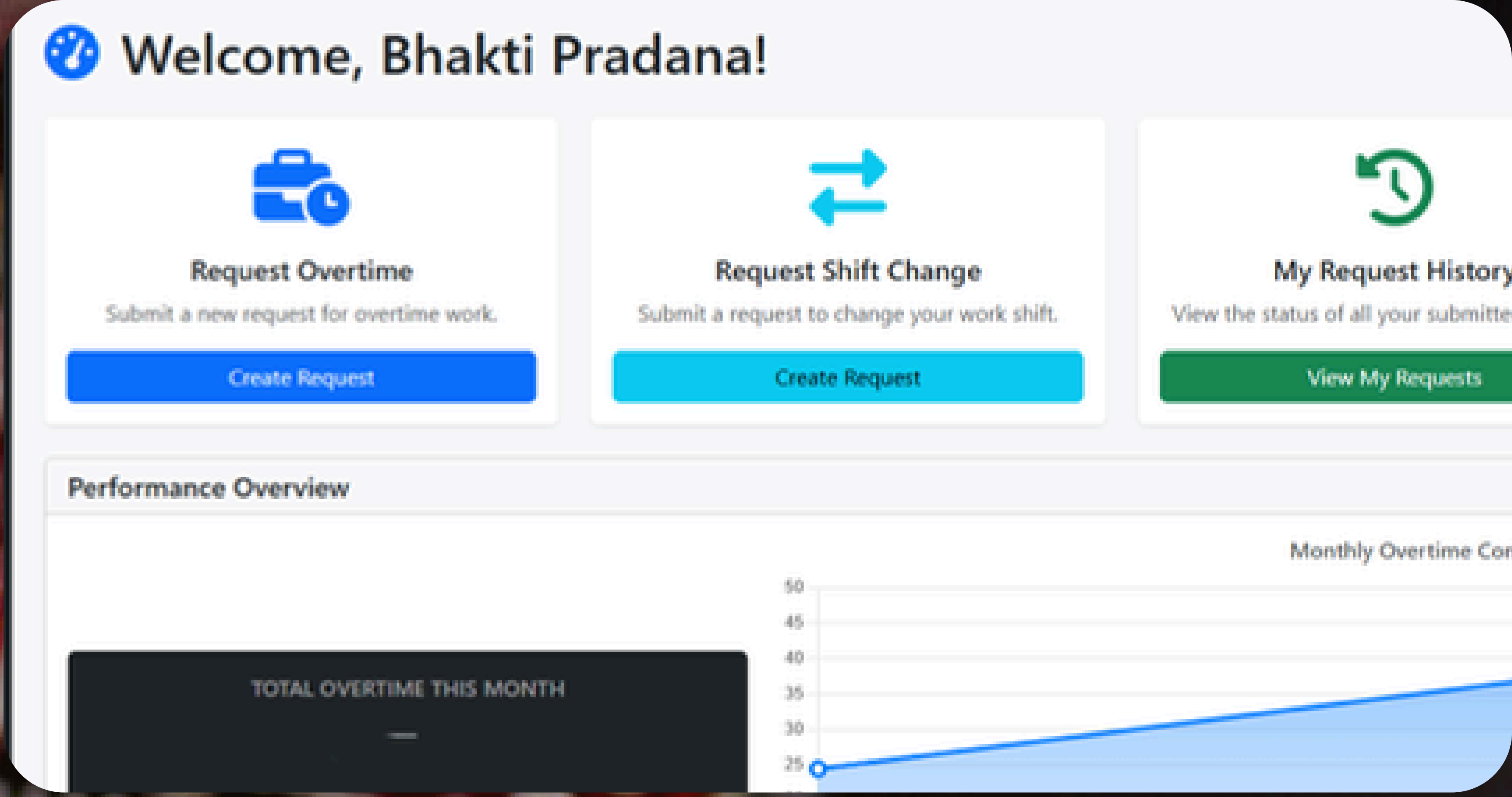
LLM Manufacturing System provides a comprehensive solution for a more efficient production process, minimizing downtime, increasing productivity, and reducing operational costs

Camera Vision



- The core function is to automatically identify various types of rejects on the PCBA based on predefined criteria, focusing on the placement, presence, and condition of components.
- The system employs state-of-the-art YOLO deep learning algorithms for object detection. YOLO is highly effective because it can quickly and accurately analyze the entire image of the PCBA to locate and classify components and potential flaws in a single pass.
- By automating the inspection process with high-precision vision, the innovation significantly minimizes the probability of miss-detection that could be made by human production operators.

OT and Shift Change Digitalization



Paperless Submission

Employees no longer need physical forms. All overtime and shift change requests are submitted digitally through the system

Employee Visibility

Employees can easily track and monitor their total accumulated overtime hours each month directly within the application, ensuring transparency and accuracy.

24/7 Accessibility

Requests for overtime or shift changes are not limited by time employees can submit them anytime and anywhere, offering significant flexibility.

Remote Approval

Managers can review and grant approval (or rejection) from anywhere and at any time, eliminating delays caused by physical location.

Semi Auto Trimming Machine

Designed and implemented a Semi-Auto Trimming Machine project that enhanced the efficiency of the trimming process through an automated rotating jig system. The system utilizes an Omron CP1E PLC to control the jig, which automatically rotates upon completion of trimming on one side, allowing seamless transition to the next side without manual intervention.

Auto Weighing Machine

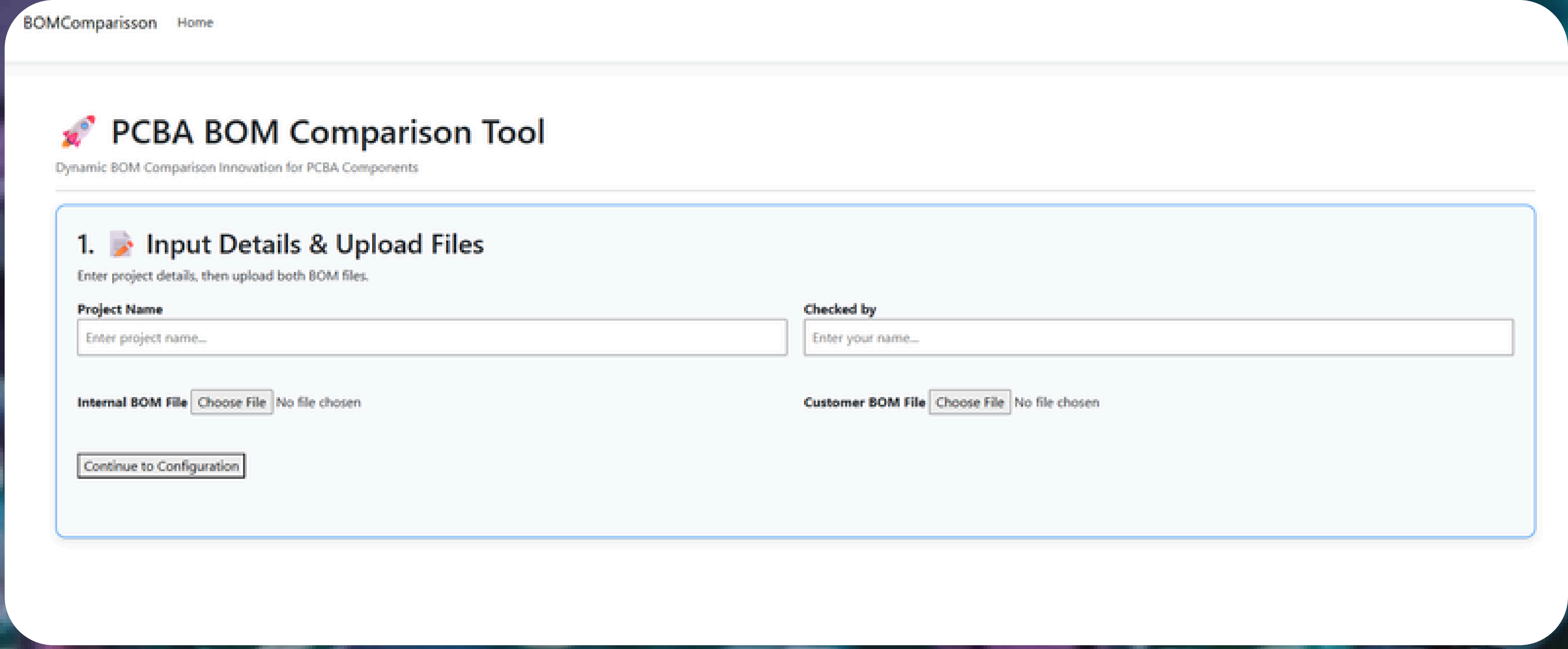
Designed and implemented an Automatic Weighing Machine as a direct response to product weight inconsistency issues and the critical need for enhanced quality control.

This project successfully overcame the limitations of manual weighing (delays and human error) with an automated system controlled by an Omron CP1E PLC. The result was improved checking accuracy, a significant increase in product output, and a reduction in manpower requirements.

NRE Tracking System

Developed and implemented an "NRE Tracking" system to streamline the monitoring and reporting of all Non-Recurring Engineering project expenditures.
This innovation provides the NPI and Process Departments with real-time visibility into cost movements, ensuring accurate weekly reporting to Management and Finance.

BOM Comparison Tool



BOM Comparison Tool: A major innovation designed to eliminate the time-consuming bottleneck of manual Bill of Materials (BOM) comparison.
Spearheaded the creation of this user-friendly web system, which automates the comparison of our Internal BOMs against diverse Customer BOM formats.Replaced hours of tedious, manual Excel work with instant and accurate data, immediately impacting the efficiency of the Program Administrator, NPI, and Process teams within the Automotive Division.

Publication
Conference and Research paper

AI Convention

"AI for Decision Makers (Europe and Asia-wide Conference)"

International Conference on Biospheric Harmony (ICOBAR)

Presenting my paper, "Leveraging Large Language Models in Industry 5.0 Manufacturing Systems for Sustainable Industrial Innovation"

International Journal of Technology

Editor Decision on #R4-IE-7557 : Accepted

Ms ID **#R4-IE-7557**
Title : Neuro-Fuzzy Inference System for Accurate Prediction of Z-Axis Values in Screw Installation
Author(s) : Bhakti Roesyadi, Taufik Roni Sahroni

Dear **Taufik Roni Sahroni** ,

Greetings from Depok,

The editorial board is delighted to inform you that your paper entitled "Neuro-Fuzzy Inference System for Accurate Prediction of Z-Axis Values in Screw Installation" has been accepted to be published on IJTech. **Congratulation!**

Paper Publication

"Neuro-Fuzzy Inference System for Accurate Prediction of Z-Axis Values in Screw Installation"

Thank You Very Much

Thank you for reviewing my innovation project portfolio. I look forward to connecting with you soon!

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